Ordinary Differential Equation

Differential equation: An equation involving derivatives of one or more dependent variables with respect to one or more independent variables is called a differential equation.

OR,

An equation involving a differential coefficient is called differential equation.

Example (i) $x^2 \frac{dy}{dx} + 2xy = 4x^2$

(ii) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = \sin x$

There are two types of differential equation:

(i) Ordinary differential equation (ODE)

(ii) Partial differential equation (PDE)

Ordinary differential equation: A differential equation involving ordinary derivatives of one or more dependent variables with respect to one independent variable is called an ordinary differential equation.

Ex: (i) $\frac{d^2y}{dx^2} + xy\left(\frac{dy}{dx}\right)^2 = 0$

(ii) $\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 + y = 0$

Partial differential equation: A differential equation involving partial derivatives of one or more dependent variables with respect to more than one independent variables is called a partial differential equation.

Example: (i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu$

(ii) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y} = 0$

(iii) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} + \frac{\partial u}{\partial t} = 0$

Order of ordinary differential equation: The order of the highest ordered derivative involved in a differential equation is called the order of the differential equation.

Example: (i) $\frac{d^3y}{dx^3} + 7\frac{d^2y}{dx^2} - 3y = x\log x$ has order 3

(ii) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$ has order 2.

Degree of ordinary differential equation: The degree of the highest ordered derivative involved in a differential equation is called the degree of the differential equation.

Ex: (i) $\frac{d^2x}{dt^2} + 2\left(\frac{dx}{dt}\right)^2 + 7x = 0$ has degree 1.

(ii) $\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{d^2y}{dx^2}\right)^8 + \frac{dy}{dx} + y = 1$ has degree 2.

Linear and Non-linear differential equation: A differential equation is called linear if

(i) it is of 1st degree in the dependent variable.

(ii) Partial derivative and product of the dependent variable and it's derivative must be absent.

Otherwise, it will be called a non-linear differential equation.

Ex: Linear Differential equation:

(i) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 2y = 0$

(ii) $x^3\frac{d^3y}{dx^3} + x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + y = x\ln x$

Non-Linear differential equation:

(i) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 3y^2 = 0$

(ii) $\frac{d^3y}{dx^3} + y\frac{dy}{dx} + y = e^x$

(iii) $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 + y = x$

Soln

Solution of differential equation:

(i) Explicit Solution

(ii) Implicit Solution

(iii) General solution

(iv) Particular solution

(v) Singular solution

Explicit solution: If the function $y = f(x)$ is the solution of a differential equation of variables x and y, then the solution is called explicit solution.

Ex: The explicit solution of $\frac{d^2y}{dx^2} = 6x$ is $y = x^3 + Ax + B$, where A, B is arbitrary constant.

Implicit solution: If the function $F(x, y) = 0$ is the solution of a differential equation of variables x and y, then it is called implicit solution.

Ex: The implicit solution of $2x\,dx + 2y\,dy = 0$ is $x^2 + y^2 = c$,

General solution: If the solution of any nth order differential equation contains n arbitrary constants, then the solution is called general solution of the differential equation.

Ex: The general solution of $\frac{dy}{dx} = y$ is $y = Ae^x$, where A is a constant.

~~i~~ Particular solution: The solution is obtained for the definite values ~~of~~ to the arbitrary constants of general solution is called particular solution of the differential equation.

Ex: The particular solution of $\frac{dy}{dx} = y$ is $y = e^x$.

Problem 1: Find the differential equation whose solution is $y = e^x(A\cos x + B\sin x)$

Solution: Given that, $y = e^x(A\cos x + B\sin x)$ --- (1)

Now differentiating both sides w.r.t. to x

$$\frac{dy}{dx} = e^x(-A\sin x + B\cos x) + e^x(A\cos x + B\sin x)$$

$$= e^x(-A\sin x + B\cos x) + y \cdots (2) \quad [\text{using } (1)]$$

Again differentiating both sides w.r.t. to x by (2)

$$\Rightarrow \frac{d^2y}{dx^2} = e^x(-A\cos x - B\sin x) + e^x(-A\sin x + B\cos x) + \frac{dy}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -e^x(A\cos x + B\sin x) + \left(\frac{dy}{dx} - y\right) + \frac{dy}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = -y + 2\frac{dy}{dx} - y$$

$$\Rightarrow \frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0,$$

which is the required differential equation.

Problem 2: Show that the differential equation of a family of circles touch the x-axis at origin is $(x^2-y^2)dy - 2xy\,dx = 0$.

Solution: The equation of the family of circles that touches x-axis at the origin is

$x^2+y^2+2fy = 0$ --- (1), where f is constant.

Now differentiating eqn (1) w.r.t. to x we get,

$2x + 2y\frac{dy}{dx} + 2f\frac{dy}{dx} = 0$

$\Rightarrow 2x + 2y\frac{dy}{dx} - \frac{x^2+y^2}{y}\frac{dy}{dx} = 0$ [From (1)]

$\Rightarrow 2xy + 2y^2\frac{dy}{dx} - (x^2+y^2)\frac{dy}{dx} = 0$

$\Rightarrow 2xy + 2y^2\frac{dy}{dx} - x^2\frac{dy}{dx} - y^2\frac{dy}{dx} = 0$

$\Rightarrow 2xy + y^2\frac{dy}{dx} - x^2\frac{dy}{dx} = 0$

$\Rightarrow (y^2-x^2)\frac{dy}{dx} + 2xy = 0$

$\Rightarrow (x^2-y^2)\frac{dy}{dx} - 2xy = 0$

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Problem 3: Show that the differential equation of a family of circles touch the y-axis at origin is $(x^2-y^2)dx + 2xy\,dy = 0$.

Solution: The equation of the family of circle that touches y-axis at the origin is,

$$x^2+y^2+2gx=0 \quad \text{---(1)}, \text{ where } g \text{ is constant.}$$

Now differentiating eqn(1) w.r.to x we have,

$$2x+2y\frac{dy}{dx}+2g=0$$

$$\Rightarrow 2x+2y\frac{dy}{dx}-\frac{x^2+y^2}{x}=0 \quad [\text{From (1)}]$$

$$\Rightarrow 2x^2+2xy\frac{dy}{dx}-(x^2+y^2)=0$$

$$\Rightarrow 2x^2dx+2xy\,dy-(x^2+y^2)dx=0$$

$$\Rightarrow 2x^2dx+2xy\,dy-x^2dx-y^2dx=0$$

$$\Rightarrow x^2dx+2xy\,dy-y^2dx=0$$

$$\Rightarrow (x^2-y^2)dx+2xy\,dy=0 \quad \text{Ans}$$

Problem 4: Find the differential equation whose solution is

$y = a + b \ln x + c(\ln x)^2 + 3x^2$.

Solution: Given that, $y = a + b\ln x + c(\ln x)^2 + 3x^2$ --- (1)

Now differentiating eqn (1) w. r. to x

$$\frac{dy}{dx} = 0 + b\frac{1}{x} + 2c\frac{\ln x}{x} + 6x$$

$$x\frac{dy}{dx} = b + 2c\ln x + 6x^2 \quad \text{---} (2)$$

Again differentiating eqn (2) w. r. to x

$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} = \frac{2c}{x} + 12x$$

$$\Rightarrow x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = 2c + 12x^2 \quad \text{---} (3)$$

Again differentiating eqn (3) w. r. to x

$$x^2\frac{d^3y}{dx^3} + 2x\frac{d^2y}{dx^2} + x\frac{d^2y}{dx^2} + \frac{dy}{dx} = 24x$$

$$\Rightarrow x^3\frac{d^3y}{dx^3} + 3x\frac{d^2y}{dx^2} + \frac{dy}{dx} - 24x = 0$$

which is the required differential equation.

Solution of 1st order and 1st degree differential equations: A differential equation of the form $M(x,y)dx + N(x,y)dy = 0$ or $\frac{dy}{dx} = f(x,y)$ is called 1st order and 1st degree differential equation.

For solving these equations we have the following methods:

(i) ~~See~~ Variable separable equation
(ii) Homogeneous equation
(iii) Exact equation
(iv) Linear equation
(v) Bernoulli and Riccati equation.

Separation of variables

Ex 1 Solve $\frac{dy}{dx}+1=y^2$

Solution: Given that, $\frac{dy}{dx}+1=y^2$

$$\Rightarrow \frac{dy}{dx}=y^2-1$$

$$\Rightarrow \int\frac{dy}{y^2-1}=\int dx$$

$$\Rightarrow \frac{1}{2\cdot 1}\ln\left|\frac{y-1}{y+1}\right|=x+c$$

$$\Rightarrow \ln\left|\frac{y-1}{y+1}\right|=2x+c$$

Ex 2: Solve $y\,dx=x\,dy$

Solution: Given that $y\,dx=x\,dy$

$$\Rightarrow \frac{dx}{x}=\frac{dy}{y}$$

$$\Rightarrow \int\frac{dx}{x}=\int\frac{dy}{y} \quad \text{[By integrating]}$$

$$\Rightarrow \ln x=\ln y+\ln c$$

$$\Rightarrow \ln x=\ln cy$$

$$\Rightarrow x=cy$$

which is the required solution.

Ex Solve $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$

Solution: Given that $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$

$$\Rightarrow \sec^2 x \tan y\, dx = -\sec^2 y \tan x\, dy$$

$$\Rightarrow \frac{\sec^2 x}{\tan x} dx = -\frac{\sec^2 y}{\tan y} dy$$

$$\Rightarrow \int \frac{\sec^2 x}{\tan x} dx + \int \frac{\sec^2 y}{\tan y} dy = 0$$

$$\Rightarrow \ln \tan x + \ln \tan y = \ln c$$

$$\Rightarrow \ln \tan x \tan y = \ln c$$

$$\Rightarrow \tan x \tan y = c$$

which is the required solution.

Ex Solve $y\sqrt{x^2-1}\, dx + x\sqrt{y^2-1}\, dy = 0$

Solution: Given that, $y\sqrt{x^2-1}\, dx + x\sqrt{y^2-1}\, dy = 0$

$$\Rightarrow y\sqrt{x^2-1}\, dx = -x\sqrt{y^2-1}\, dy$$

$$\Rightarrow \frac{\sqrt{x^2-1}}{x} dx = -\frac{\sqrt{y^2-1}}{y} dy$$

$$\Rightarrow \frac{x^2-1}{x\sqrt{x^2-1}} dx = -\frac{(y^2-1)}{y\sqrt{y^2-1}} dy$$

$$\Rightarrow \frac{x^2-1}{x\sqrt{x^2-1}} dx + \frac{y^2-1}{y\sqrt{y^2-1}} dy = 0$$

$$\Rightarrow \frac{x^2}{x\sqrt{x^2-1}} dx - \frac{dx}{x\sqrt{x^2-1}} + \frac{y^2}{y\sqrt{y^2-1}} dy - \frac{dy}{y\sqrt{y^2-1}} = 0.$$

$$\Rightarrow \frac{x}{\sqrt{x^2-1}}dx + \frac{y}{\sqrt{y^2-1}}dy = \frac{dx}{x\sqrt{x^2-1}} + \frac{dy}{y\sqrt{y^2-1}}$$

$\Rightarrow$ By integrating on both sides,

$$\Rightarrow \int \frac{x}{\sqrt{x^2-1}}dx + \int \frac{y}{\sqrt{y^2-1}}dy = \int \frac{dx}{x\sqrt{x^2-1}} + \int \frac{dy}{y\sqrt{y^2-1}}$$

$$\Rightarrow \cancel{\int \frac{1}{2}\frac{2x}{\sqrt{x^2-1}}dx + \int \frac{1}{2}\frac{2y}{\sqrt{y^2-1}}dy = \int \frac{dx}{x\sqrt{x^2-1}} + \int \frac{dy}{y\sqrt{y^2-1}}}$$

$$\Rightarrow \int \frac{\frac{1}{2}d(x^2-1)}{\sqrt{x^2-1}} + \int \frac{\frac{1}{2}d(y^2-1)}{\sqrt{y^2-1}} = \sec^{-1}x + \sec^{-1}y + c$$

$$\Rightarrow \frac{1}{2}\frac{(x^2-1)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + \frac{1}{2}\frac{(y^2-1)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \sec^{-1}x + \sec^{-1}y + c$$

$$\Rightarrow \sqrt{x^2-1} + \sqrt{y^2-1} = \sec^{-1}x + \sec^{-1}y + c$$

which is the required solution.

Solve $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$.

Soln: Given that, $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$

$$\Rightarrow \frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$$

$$\Rightarrow \frac{dy}{\sqrt{1-y^2}} = \frac{-dx}{\sqrt{1-x^2}}$$

$$\Rightarrow \int \frac{dx}{\sqrt{1+x^2}} + \int \frac{dy}{\sqrt{1-y^2}} = 0$$

$$\Rightarrow \sin^{-1}x + \sin^{-1}y = c$$

which is the required solution.

Solve $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

Solution: Given that, $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

$\Rightarrow \frac{dy}{dx} = e^x . e^{-y} + x^2 e^{-y}$

$\Rightarrow \frac{dy}{dx} = e^{-y}(e^x + x^2)$

$\Rightarrow \frac{dy}{e^{-y}} = (e^x + x^2) dx$

$\Rightarrow e^y dy = (e^x + x^2) dx$

$\Rightarrow \int e^y dy = \int (e^x + x^2) dx$

$\Rightarrow e^y = e^x + \frac{x^3}{3} + c$ Ans